

Lec 45

Given an array of ⁿ elements, to count number of inversions what will be worst case time complexity

$a[n] \Rightarrow$ an array

algo

Inversion

program

Best Case

n^2
 $n \log n$
 Θ

Solution:

- What is inversion ?

$a[n]$ $[0 \ 1 \ 2 \ \dots \ n-1]$

let two index i , and j
let $i < j$ and $a[i] > a[j]$

This is one

inversion

$i < j$
 but $a[i] > a[j]$

$count = 0 + 7 = 3$

$count = 3$

- Ex:

$a[5] \Rightarrow [3, 2, 1, 6, 7]$
 $0 \ 1 \ 2 \ 3 \ 4$

value = 3

$a[0] \Rightarrow$ index = 0,
 $a[0]$ and $a[1]$

$0 < 1$
 $a[0] > a[1]$

$\Rightarrow \underline{\underline{O(n^2)}}$

$count = 0$

```
for (i = 0; i < n; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (a[i] > a[j])
            count++;
    }
}
```

This is Brute force approach

if $i = 0$, $j = 1, 2, 3, 4, 5$
 if $i = 1$, $j = 2, 3, 4, 5$

Algo: brute force Approach

count = 0;

$T.C = O(n^2)$

```
for (i = 0; i < n; i++)  
{  
    for (j = i + 1; j < n; j++)  
    {  
        if (a[i] > a[j])  
        {  
            count++  
        }  
    }  
}
```

Ans = 12

count = 0
3
5
6
10
12

Ex: How many inversions

a = [6, 3, 5, 4, 1, 2]

count = 12

$O(n^2)$

$$n^2 \rightarrow \underline{\underline{n \log n}}$$

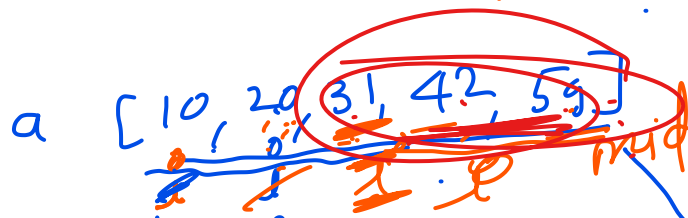
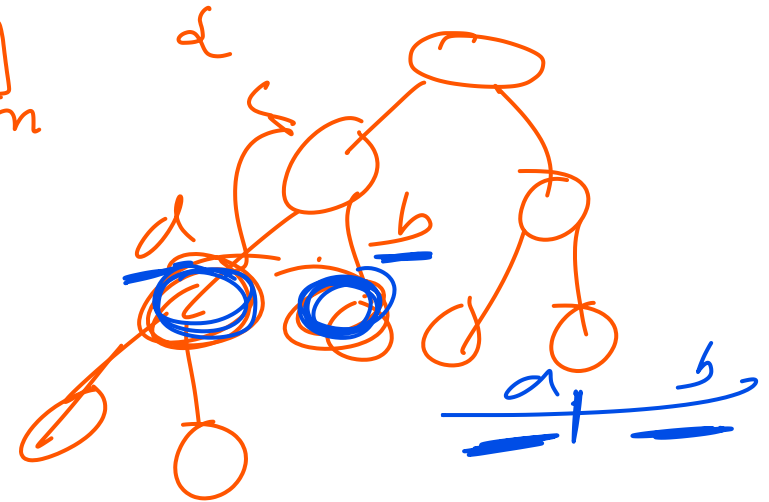
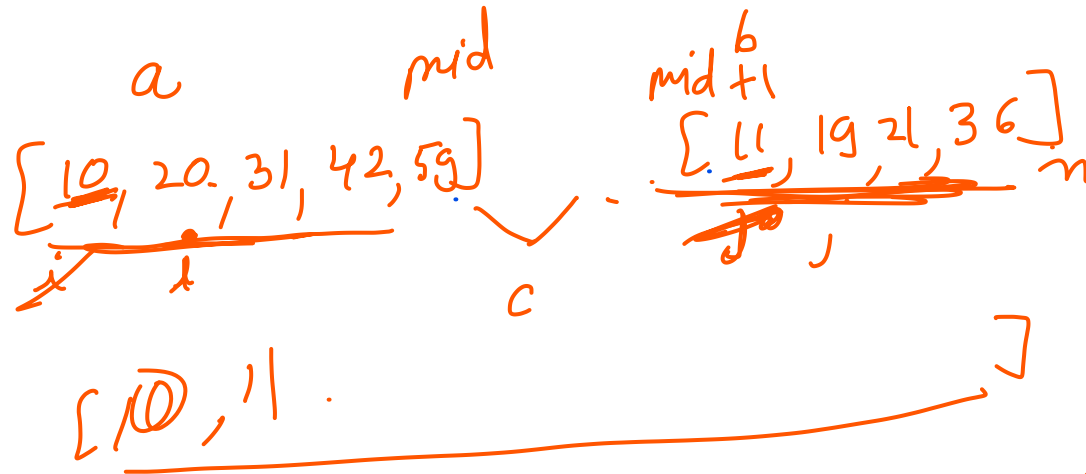
$$Tc = n \log n$$

$$\underline{\underline{O(n^2)}} \Rightarrow \text{modify}$$

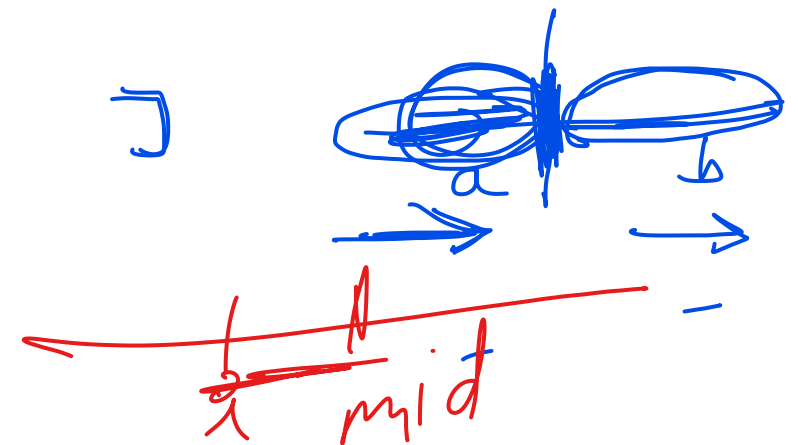
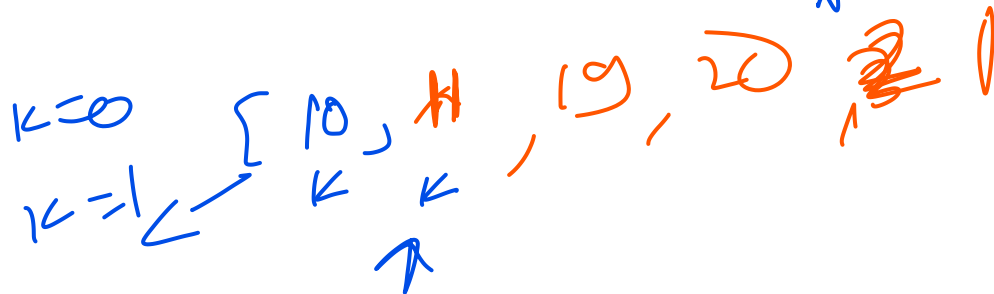
Using Divide and Conquer (Modified Merge Sort)

• Ex:

while (l <= mid
and j <= n-1)



Index of array a will be, lesser than index of array



Algo:

let no of inversions $\Rightarrow ni$, $k=0$

while ($i \leq mid$ and $j \leq n-1$)

if ($a[i] > b[j]$)

{

$c[k] = a[i]$

$k++$, $i++$

$ni = ni + mid - i + 1$

}

if ($a[i] < b[j]$)

{

$c[k] = a[i]$

$i++$, $k++$

}

}

copy remaining elements of a and b into c

$\Rightarrow$ Inver added in merged

mergesort(a, i, j)

mid = $\frac{i+j}{2}$

mergesort(a, i, mid)

mergesort(a, mid+1, j)

merge(a, i, mid, j)

$n/2$

$n/2$

$\Rightarrow O(n)$

1

$$T(n) = 2T(n/2) + \underline{n}$$

$$\Rightarrow n \log n$$

